

Continuous Assessment V

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Question 1

0.0.1 (Magnetic susceptibility behaviour for weak magnetic fields)

We have the Hamiltonian

$$E = \frac{p^2}{2m} - \sigma \mu_B B \quad (1)$$

where we can define $\sigma = \pm 1$ for $\uparrow, \downarrow$.

$$\Xi = \prod_{i=1}^N \zeta_i = \zeta_i^N = \left[1 + e^{-\beta \left(\frac{p_i^2}{2m} - \sigma \mu_B B - \mu \right)} \right] \quad (2)$$

$$\Rightarrow \ln \Xi = - \sum_{i=1}^N \ln \left(1 + e^{-\beta \left(\frac{p_i^2}{2m} - \sigma \mu_B B - \mu \right)} \right) \text{ let } z_{\pm} = e^{\beta(\mu \pm \mu_B H)} \quad (3)$$

$$\Rightarrow \ln \Xi = \sum_{i=1}^N \left[\ln (1 + e^{-\beta \epsilon_n z_+}) + \ln (1 + e^{-\beta \epsilon_n z_-}) \right] \quad (4)$$

$$(5)$$

We can treat this semi-classically

$$\ln \Xi = - \int_0^\infty W(\epsilon) \left[\ln (1 + e^{-\beta \epsilon_n z_+}) + \ln (1 + e^{-\beta \epsilon_n z_-}) \right] \quad (6)$$

$$W(\epsilon) = \delta(\epsilon) + \sum_{i \neq 0} \delta(\epsilon - \epsilon_i) \approx (\text{deg}) \int_{-\infty}^\infty d^3 \delta \left(\epsilon - \frac{h^2 i^2}{2mL^2} \right) \quad (7)$$

$$\approx (\text{deg}) \int_{-\infty}^\infty \frac{L^3}{h^3} dp^3 \delta \left(\epsilon - \frac{p^2}{2m} \right) \quad (8)$$

$$\approx 4\pi (\text{deg}) \frac{L^3}{h^3} \int_0^\infty p^2 dp \delta \left(\epsilon - \frac{p^2}{2m} \right) \approx \frac{8\pi V m^2}{h^3 \sqrt{2m}} (\text{deg}) \frac{L^3}{h^3} \int_0^\infty \sqrt{u} dp \delta(\epsilon - u) \quad (9)$$

$$W(\epsilon) \approx \frac{8\pi V m^2}{h^3 \sqrt{2m}} (\text{deg}) \frac{L^3}{h^3} \sqrt{\epsilon} \quad (10)$$

$$\text{Let } \mathcal{A} = \frac{8\pi V m^2}{h^3 \sqrt{2m}} (\text{deg}) \frac{L^3}{h^3} \quad (11)$$

$$\ln \Xi \approx - \int_0^\infty \mathcal{A} \sqrt{\epsilon} \ln (1 + e^{-\beta \epsilon_n z_{\pm}}) = \left[\ln (1 + e^{-\beta \epsilon} z_{\pm}) \left(\mathcal{A} \frac{2}{3} \epsilon^{\frac{3}{2}} \right) \right] \Big|_0^\infty - \int_0^\infty \mathcal{A} \frac{2}{3} \epsilon^{\frac{3}{2}} \left(\frac{e^{-\beta \epsilon} (-\beta z_{\pm})}{1 + e^{-\beta \epsilon} z_{\pm}} d\epsilon \right) \quad (12)$$

$$- \int_0^\infty \mathcal{A} \frac{2}{3} \epsilon^{\frac{3}{2}} \left(\frac{e^{-\beta \epsilon} (-\beta z_{\pm})}{1 + e^{-\beta \epsilon} z_{\pm}} d\epsilon \right) \quad (13)$$

$$- \frac{1}{\beta^{\frac{3}{2}}} \int_0^\infty \mathcal{A} \frac{3}{2} (\beta \epsilon)^{\frac{3}{2}} \left(\frac{1}{-1 + \frac{e^{\beta \epsilon}}{-z_{\pm}}} d(\beta \epsilon) \right) \quad (14)$$

$$= - \frac{2}{3} \mathcal{A} \int_0^\infty \frac{\epsilon^{\frac{3}{2}}}{\frac{e^{\beta \epsilon}}{-\beta z_{\pm}} - 1} d\epsilon = - \frac{2}{3} \Gamma \left(\frac{1}{2} \right) \mathcal{A} \mathcal{L} i_{\frac{5}{2}}(-z_{\pm}) = - \frac{2}{3} \mathcal{A} \Gamma \left(\frac{1}{2} \right) \frac{1}{\beta^{\frac{3}{2}}} \left[\mathcal{L} i_{\frac{5}{2}}(-z_+) + \mathcal{L} i_{\frac{5}{2}}(-z_-) \right] \quad (15)$$

$$\ln \Xi = - (\text{deg}) \frac{V}{\lambda_T^3} \left[\mathcal{L} i_{\frac{5}{2}}(-z_+) + \mathcal{L} i_{\frac{5}{2}}(-z_-) \right] \quad (16)$$

We know $J = -kT \ln \Xi$ and we have seen in lectures that

$$J = (\text{deg}) \frac{kTV}{\lambda_T^3} \left[\mathcal{L}i_{\frac{5}{2}}(-z_+) + \mathcal{L}i_{\frac{5}{2}}(-z_-) \right] \text{ with } \lambda_T = \sqrt{\frac{2\pi\hbar^2}{mkT}} \propto T^{-\frac{1}{2}} \quad (17)$$

The average particle number is given by

$$\langle N \rangle = z \frac{\partial \ln \Xi}{\partial z} = z_+ \frac{\partial \ln \Xi}{\partial z_+} + z_- \frac{\partial \ln \Xi}{\partial z_-} \quad (18)$$

$$= -(\text{deg}) \frac{V}{\lambda_T^3} \left[\mathcal{L}i_{\frac{3}{2}}(-z_+) + \mathcal{L}i_{\frac{3}{2}}(-z_-) \right] \quad (19)$$

The average magnetization is given by

$$\langle M \rangle = -\frac{1}{N} \frac{\partial J}{\partial H} = -\frac{(\text{deg})kTV}{N} \left[\frac{\partial \mathcal{L}i_{\frac{5}{2}}(-z_+)}{\partial H} + \frac{\partial \mathcal{L}i_{\frac{5}{2}}(-z_-)}{\partial H} \right] = \dots \left[\frac{\partial \mathcal{L}i_{\frac{5}{2}}(-z_+)}{\partial H} \frac{\partial(-z_+)}{\partial H} + \frac{\partial \mathcal{L}i_{\frac{5}{2}}(-z_-)}{\partial H} \frac{\partial(-z_-)}{\partial H} \right] \quad (20)$$

$$\frac{\partial J}{\partial H} = -(\text{deg}) \cdot \frac{kTV\mu_B}{\lambda_T^3} \left[\mathcal{L}i_{3/2}(-z_+) - \mathcal{L}i_{3/2}(-z_-) \right] \quad (21)$$

$$\langle M \rangle = (\text{deg}) \frac{kTV\mu_B}{N\lambda_T^3} \left[\mathcal{L}i_{3/2}(-z_+) - \mathcal{L}i_{3/2}(-z_-) \right] = \quad (22)$$

$$\langle M \rangle = -\frac{1}{N} \frac{\partial J}{\partial H} = \underbrace{\frac{\mu_B}{N} \frac{V}{\lambda_T^3}}_{\alpha} \left[\mathcal{L}i_{\frac{3}{2}}(-z_+) - \mathcal{L}i_{\frac{3}{2}}(-z_-) \right] \quad (23)$$

The magnetic susceptibility is given by

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_{T,V,N} = \alpha \left[\left(\frac{\partial(-z_+)}{\partial H} \right) \frac{1}{-z_+} \mathcal{L}i_{\frac{1}{2}}(-z_+) - \left(\frac{\partial(-z_-)}{\partial H} \right) \frac{1}{-z_-} \mathcal{L}i_{\frac{1}{2}}(-z_-) \right] \quad (24)$$

$$= \alpha \left[(-\mu_B \beta Z_+) \frac{1}{-z_+} \mathcal{L}i_{\frac{1}{2}}(-z_+) - (\mu_B \beta Z_-) \frac{1}{-z_-} \mathcal{L}i_{\frac{1}{2}}(-z_-) \right] \quad (25)$$

$$\chi_T = -\frac{\mu_B}{N} \frac{V}{\lambda_T^3} (\beta \mu_T) \left[\mathcal{L}i_{\frac{1}{2}}(-z_+) + \mathcal{L}i_{\frac{1}{2}}(-z_-) \right] \quad (26)$$

(Adherence to the Curie Law, and independence of T)

In the weak field limit ($\mu_B H \ll kT$) we have:

$$\langle N \rangle = -(\text{deg}) \frac{2V}{\lambda_T^3} \left[\mathcal{L}i_{\frac{3}{2}}(-z) \right] \quad (27)$$

$$z_{\pm} \rightarrow z \quad (28)$$

$$\chi_T = -\frac{2\mu_B}{N} \frac{V}{\lambda_T^3} (\beta \mu_T) \left[\mathcal{L}i_{\frac{1}{2}}(-z) \right] \quad (29)$$

negligible alignment.

In the high temperature regime $\beta \ll 1$

$$\mathcal{L}i_{\frac{1}{2}}(-z) \approx -z + \dots \quad (30)$$

$$\langle N \rangle \underset{z \ll 1}{=} \frac{2Vz}{\lambda_T^3} \implies \chi_T = \frac{V\beta\mu_B^2(-2z)}{\lambda_T^3} = \beta\mu_B \implies \chi_T \propto T^{-1} \quad (31)$$

We see the manifestation of the curie law.

In the low temperature regime $\beta \gg 1$

$$\mathcal{L}i_{\frac{1}{2}}(-z) \underset{z \gg 1}{\approx} \frac{(\ln z)^{\frac{1}{2}}}{\Gamma(\frac{3}{2})} \quad (32)$$

$$\langle N \rangle = -2(\deg) \frac{2V}{\lambda_T^3} \left[\frac{(\ln z)^{\frac{3}{2}}}{\Gamma(\frac{3}{2})} \right] \quad (33)$$

$$\chi_T = -\frac{2\mu_B}{N} \frac{V}{\lambda_T^3} (\beta\mu_B) \left[\frac{(\ln z)^{\frac{1}{2}}}{\Gamma(\frac{3}{2})} \right] = \frac{3}{2} \frac{\beta\mu_B^2}{\ln z} = \frac{3}{2} \frac{\mu_B^2}{\mu} \underset{T \ll 1}{=} \frac{3}{2} \frac{\mu_B^2}{\epsilon_F} \quad (34)$$

Clearly there is no temperature dependence.

Question 2

Generally the Hamiltonian for the 1D Ising model is given below. For this question we set $H = 0$. **Note I mis-labeled the magnetic field throughout and called it μ , I'm sorry.**

$$\mathcal{H} = -J \sum_i s_i s_{i+1} - H \sum_i s_i \quad (35)$$

$$\langle s_i \rangle = \frac{1}{Z} \text{Tr} \left(\sigma_z \mathcal{T}^{N-1} \right), \text{ with } \mathcal{T} = \begin{bmatrix} e^{\beta J + \beta \mu} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J - \beta \mu} \end{bmatrix}, \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (36)$$

$$\mathcal{T}_{diag} = \mathcal{S}^{-1} \mathcal{T} \mathcal{S} \text{ and } \mathcal{T}^{N-1} = \underbrace{\mathcal{S}^{-1} \mathcal{T} \mathcal{S} \dots \mathcal{S}^{-1} \mathcal{T} \mathcal{S}}_{N-1 \text{ times}} = \mathcal{S} \mathcal{T}^{N-1} \mathcal{S}^{-1} \quad (37)$$

$$\text{Tr} \left(\sigma_z \mathcal{T}^{N-1} \right) = \text{Tr} \left(\sigma_z \mathcal{S} \mathcal{T}_{diag}^{N-1} \mathcal{S}^{-1} \right) = \text{Tr} \left(\overline{\sigma_z} \mathcal{T}_{diag}^{N-1} \right) \text{ where } \overline{\sigma_z} = \mathcal{S} \sigma_z \mathcal{S}^{-1} \quad (38)$$

To find $\mathcal{S}$ we need the eigenvectors

$$\mathcal{T} - \lambda \mathcal{I} = \begin{bmatrix} e^{\beta J + \beta \mu} - \lambda & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J - \beta \mu} - \lambda \end{bmatrix} \quad (39)$$

let $a = e^{\beta J}$, $x = \beta \mu$, then we get the characteristic equation.

$$(ae^x - \lambda)(ae^{-x} - \lambda) - a^{-2} = 0 \quad (40)$$

$$a^2 - ae^x \lambda - ae^{-x} \lambda + \lambda^2 - a^{-2} = 0 \quad (41)$$

$$\lambda^2 - a\lambda(e^x + e^{-x}) + a^2 - a^{-2} = 0 \quad (42)$$

$$\lambda^2 - 2a\lambda \cosh(x) + a^2 - a^{-2} = 0 \quad (43)$$

$$\lambda_{\pm} = a \cosh(x) \pm \sqrt{\cosh^2(x) - (a^2 - a^{-2})} \quad (44)$$

$$\lambda_{\pm} = e^{\beta J} \left[\cosh(\beta \mu) \pm \sqrt{\sinh^2(\beta \mu) + e^{-4\beta J}} \right] \quad (45)$$

We can now find the eigenvectors

$$(\mathcal{T} - \lambda I) v_{\pm} = 0 \quad (46)$$

$$\begin{bmatrix} e^{\beta J + \beta \mu} - \lambda_{\pm} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J - \beta \mu} - \lambda_{\pm} \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \quad (47)$$

$$(e^{\beta J + \beta \mu} - \lambda_{\pm}) x + e^{-\beta J} y = 0 \quad (48)$$

$$y = e^{\beta J} (\lambda_{\pm} - e^{\beta J + \beta \mu}) x = 0 \quad (49)$$

$$\Rightarrow v_{\pm} = \frac{1}{\sqrt{1 + [(\lambda_{\pm} - e^{\beta J + \beta \mu}) e^{\beta J}]^2}} \begin{pmatrix} 1 \\ (\lambda_{\pm} - e^{\beta J + \beta \mu}) e^{\beta J} \end{pmatrix} \quad (50)$$

$$\mathcal{S} = [v_+ v_-] \text{ if we call } \alpha_{\pm} = (\lambda_{\pm} - e^{\beta J + \beta \mu}) e^{\beta J} \quad (51)$$

$$\Rightarrow \mathcal{S} = \begin{bmatrix} \frac{1}{\sqrt{1 + \alpha_+^2}} & \frac{1}{\sqrt{1 + \alpha_-^2}} \\ \frac{\alpha_+}{\sqrt{1 + \alpha_+^2}} & \frac{\alpha_-}{\sqrt{1 + \alpha_-^2}} \end{bmatrix} \quad (52)$$

For a 2x2 matrix we have the inverse given by

$$\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \Rightarrow ad - bc = \frac{\alpha_-}{\sqrt{(1 + \alpha_+^2)(1 + \alpha_-^2)}} - \frac{\alpha_+}{\sqrt{(1 + \alpha_+^2)(1 + \alpha_-^2)}} = \frac{\alpha_- - \alpha_+}{\sqrt{(1 + \alpha_+^2)(1 + \alpha_-^2)}}. \quad (53)$$

$$\Rightarrow \mathcal{S}^{-1} = \frac{\sqrt{(1 + \alpha_+^2)(1 + \alpha_-^2)}}{\alpha_- - \alpha_+} \begin{bmatrix} \frac{\alpha_-}{\sqrt{1 + \alpha_-^2}} & -\frac{1}{\sqrt{1 + \alpha_-^2}} \\ -\frac{\alpha_+}{\sqrt{1 + \alpha_+^2}} & \frac{1}{\sqrt{1 + \alpha_+^2}} \end{bmatrix}. \quad (54)$$

(H=0)

In this case

$$\lambda_{\pm} = e^{\beta J} [1 \pm e^{-2\beta J}] = e^{\beta J} \pm e^{-\beta J} = \begin{cases} 2 \cosh(\beta J) & \text{if } \lambda_+ \\ 2 \sinh(\beta J) & \text{if } \lambda_- \end{cases} \quad (55)$$

$$\Rightarrow \alpha_{\pm} = \begin{cases} (2 \cosh(\beta J) - e^{\beta J}) e^{\beta J} & \text{if } \lambda_+ \\ (2 \sinh(\beta J) - e^{\beta J}) e^{\beta J} & \text{if } \lambda_- \end{cases} \Rightarrow \alpha_{\pm} = \begin{cases} 1 & \text{if } \lambda_+ \\ -1 & \text{if } \lambda_- \end{cases} \quad (56)$$

$$\Rightarrow \mathcal{S} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \text{ and we find } \mathcal{S}^{-1} = \mathcal{S} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (57)$$

$$\overline{\sigma}_z = \mathcal{S} \sigma_z \mathcal{S}^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (58)$$

$$\langle s_i \rangle = \frac{1}{Z} \text{Tr} \left(\overline{\sigma}_z \mathcal{T}_{diag}^{N-1} \right) = 0 \quad (59)$$

We have a product of a traceless matrix and a diagonal matrix which must be traceless so the expected spin vanishes.

Question 3

0.0.2 (Expected Spin w/ Magnetic field w/ $N \gg 1$)

$$\mathcal{T} = \mathcal{S} \mathcal{T}_{diag} \mathcal{S}^{-1}, \quad \mathcal{T}_{diag} = \begin{bmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{bmatrix},$$

so that

$$\mathcal{T}^{N-1} = (\mathcal{S} \mathcal{T}_{\text{diag}} \mathcal{S}^{-1})^{N-1} = \mathcal{S} \begin{bmatrix} \lambda_+^{N-1} & 0 \\ 0 & \lambda_-^{N-1} \end{bmatrix} \mathcal{S}^{-1}.$$

Then

$$\langle s_i \rangle = \frac{1}{Z} \text{Tr}(\sigma_z \mathcal{T}^{N-1}) = \frac{\text{Tr}(\sigma_z \mathcal{S} \mathcal{T}_{\text{diag}}^{N-1} \mathcal{S}^{-1})}{\text{Tr}(\mathcal{T}^{N-1})} = \frac{\text{Tr}(\overline{\sigma}_z \mathcal{T}_{\text{diag}}^{N-1})}{\text{Tr}(\mathcal{T}_{\text{diag}}^{N-1})}, \quad \overline{\sigma}_z = \mathcal{S}^{-1} \sigma_z \mathcal{S}.$$

$$\text{Tr}(\mathcal{T}^{N-1}) = \lambda_+^{N-1} + \lambda_-^{N-1}, \quad \text{Tr}(\sigma_z \mathcal{T}^{N-1}) = \lambda_+^{N-1} \overline{\sigma}_{z11} + \lambda_-^{N-1} \overline{\sigma}_{z22},$$

$$\langle s_i \rangle = \frac{\lambda_+^{N-1} \overline{\sigma}_{z11} + \lambda_-^{N-1} \overline{\sigma}_{z22}}{\lambda_+^{N-1} + \lambda_-^{N-1}} = \frac{\overline{\sigma}_{z11} + \left(\frac{\lambda_-^{N-1}}{\lambda_+^{N-1}} \right) \overline{\sigma}_{z22}}{1 + \frac{\lambda_-^{N-1}}{\lambda_+^{N-1}}} \quad (60)$$

In the thermodynamic limit $N \gg 1$, if $|\lambda_+| > |\lambda_-|$, the largest eigenvalue λ_+ dominates, giving so

$$\langle s_i \rangle \approx \overline{\sigma}_{z11},$$

$$\overline{\sigma}_z = \begin{bmatrix} \frac{1}{\sqrt{1+\alpha_+^2}} & \frac{1}{\sqrt{1+\alpha_-^2}} \\ \frac{\alpha_+}{\sqrt{1+\alpha_+^2}} & \frac{\alpha_-}{\sqrt{1+\alpha_-^2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{\sqrt{(1+\alpha_+^2)(1+\alpha_-^2)}}{\alpha_- - \alpha_+} \begin{bmatrix} \frac{\alpha_-}{\sqrt{1+\alpha_-^2}} & -\frac{1}{\sqrt{1+\alpha_-^2}} \\ -\frac{\alpha_+}{\sqrt{1+\alpha_+^2}} & \frac{1}{\sqrt{1+\alpha_+^2}} \end{bmatrix}. \quad (61)$$

$$\overline{\sigma}_z = \frac{1}{\alpha_- - \alpha_+} \begin{bmatrix} \alpha_- + \alpha_+ & -2 \\ 2\alpha_+ \alpha_- & -(\alpha_- + \alpha_+) \end{bmatrix} \implies \overline{\sigma}_{z11} = \frac{\alpha_- + \alpha_+}{\alpha_- - \alpha_+} \quad (62)$$

$$\overline{\sigma}_{z11} = \frac{\lambda_- + \lambda_+ - 2e^{\beta J + \beta \mu}}{\lambda_- - \lambda_+} = \quad (63)$$

where v_+ is the eigenvector associated with λ_+ .

$$\overline{\sigma}_{z11} = \frac{e^{\beta J} [2 \cosh(\beta \mu) - 2e^{\beta \mu}]}{e^{\beta J} [-2\sqrt{\sinh^2(\beta \mu)} + e^{-4\beta J}]} = \frac{\sinh(\beta \mu)}{\sqrt{\sinh^2(\beta \mu) + e^{-4\beta J}}} \quad (64)$$

$(\beta H \gg 1)$

What is the value of $\langle s_i \rangle$ and what does it indicate?

We have

$$\overline{\sigma_{z11}} = \frac{\sinh(\beta\mu)}{\sqrt{\sinh^2(\beta\mu) + e^{-4\beta J}}} = \frac{\frac{e^{\beta\mu} - e^{-\beta\mu}}{2}}{\sqrt{\left(\frac{e^{\beta\mu} - e^{-\beta\mu}}{2}\right)^2 + e^{-4\beta J}}} \quad (65)$$

$$(66)$$

If we have $\beta H \gg 1$ for $H > 0$

$$\frac{\frac{e^{\beta\mu} - e^{-\beta\mu}}{2}}{\sqrt{\left(\frac{e^{\beta\mu} - e^{-\beta\mu}}{2}\right)^2 + e^{-4\beta J}}} \rightarrow \frac{\frac{e^{\beta\mu}}{2}}{\sqrt{\left(\frac{e^{\beta\mu}}{2}\right)^2 + e^{-4\beta J}}} \rightarrow \frac{e^{\beta\mu}}{e^{\beta\mu}} = 1 \quad (67)$$

For the $H < 0$ case we have the $-e^{\beta\mu}$ terms surviving so we get -1 .

Thus in the $\beta H \gg 1$ regime we have

$$\langle s_i \rangle \Big|_{H>0} = 1, \quad \langle s_i \rangle \Big|_{H<0} = -1 \quad (68)$$

Where for $+1$ they are perfectly aligned and -1 they are perfectly misaligned.

Question 4

0.0.3 (Average spin without magnetic field under $\beta J \rightarrow \infty$)

To recover the expected result one should let the magnetic field vanish then take the limit then . The initial presence of the magnetic field brakes the spin symmetry

$$\langle s_i \rangle = \frac{\sinh(\beta\mu)}{\sqrt{\sinh^2(\beta\mu) + e^{-4\beta J}}} \quad (69)$$

If we let the magnetic field vanish then take the limit

$$\langle s_i \rangle = 0 \quad (70)$$

as the numerator vanishes before we can apply the limit.

If we take the limit then let the field vanish

$$\langle s_i \rangle = \frac{\sinh(\beta\mu)}{\sqrt{\sinh^2(\beta\mu) + e^{-4\beta J}}} \rightarrow \frac{\sinh(\beta\mu)}{\sqrt{\sinh^2(\beta\mu)}} \rightarrow \frac{\sin(\beta\mu)}{|\sin(\beta\mu)|} = \pm 1 \quad (71)$$

where the sign depends of if we take $\lim_{\mu \rightarrow 0^+}$ or $\lim_{\mu \rightarrow 0^-}$. We a non analytic expression for $\beta\mu = 0$ indicating a phase transition.

Thus to get the expected result we take the limit $\beta J \gg 1$ the magnetic field vanish .